Predicting Hospital Readmissions for Diabetic Patients

Problem Statement

Readmissions to hospitals continue to be a major problem in healthcare which could be due to various diseases, inadequate treatment, medical errors, early discharge, and other socioeconomic factors. Identifying patients at a high risk of readmission help hospitals allocate resources efficiently, reduce unnecessary readmissions and costs. In today's world, Diabetes, a chronic condition, has become very common across all age groups which accounts to higher readmission rates due to complications and poor management. This project is to predict hospital readmissions for diabetic patients using a large dataset that includes demographic data, medical history and other clinical features, focusing on whether factors like glucose levels, A1C results, and medication usage are key predictors of readmission.

Dataset Description

This projects uses the [Diabetic Patients' Re-admission Prediction]() dataset containing 101766 rows of data with 50 features which was sourced from Kaggle.
It includes features such as patient demographics (race, gender, age, weight), hospitalization details (admission type, discharge disposition, admission source), medical history (diagnoses, number of lab procedures, medications), and treatment details (insulin, diabetes medications, A1C results). The dependent variable, 'readmitted,' has 3 outcomes: <30 (admitted within 30 days of discharge), >30 (admitted after 30 days of discharge) and NO (no readmission). For this project, I will be combining (<30, >30) into a single outcome **YES(readmitted)**. The target variable now has two outcomes Yes and No.

Machine Learning Approch

1. Data Preprocessing

- Missing values like **payer_code, weight, medical_specialty** must be handled and unnecessary variables like **encounter_id, patient_nbr** must be dropped.
- Categorical features like **race and gender** are encoded.
- Numerical features like **time_in_hospital** and **num_lab_procedures** are normalized.
- The key features are visualized and correlations between them are identified.

2. Feature Engineering

- New variables like **total_visits** are created by aggregating the number of inpatient, outpatient and emergency visits.
- Oversampling/undersampling techniques are used to address the class imbalances.
- Feature selection techniques are used to improve the model performance.

3. Model Training

- The data is split into test and training sets and then the data is trained on various machine learning models like **Logistic Regression, Decision Trees** and **Random Forests**.
- It is then compared by training the ensemble models like **XGBoost**.
- **Cross-validation** is performed to evaluate the models and prevent overfitting. Also, hyperparameters can be tuned to optimize the performance.

4. Model Evaluation

- The model is evaluated using various metrics like **Accuracy, Precision, Recall, F1-Score** and **AUC-ROC**.
- The key factors predicting the readmissions can be identified by using the feature importance scores.

References

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